Medical Device Regulatory Dashboard Project Report

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Source workbook Shraddha_Dashboard_FINAL2.xlsx
Report date 5 November 2026
Recorded approval years 2019 to 2025

Executive summary

This project uses Excel to organise 200 medical device records and summarise regulatory status, complaint counts, adverse events, recall flags and geographic coverage. The workbook brings together 15 device types, four class labels and ten countries, with dropdown controls for targeted review.

Measure	Calculated result
Device records	200
Records marked Approved	56 / 28.00%
Records flagged Recall Required	110 / 55.00%
Reported complaints	15,488
Reported adverse events	3,898
Average adverse events per record	19.49
UDI Assigned Yes	101 / 50.50%
PostMarket Study Required	106 / 53.00%

Source: Independently calculated from Tables!A2:A201, H2:H201 and J2:N201, with all records included. Formula-cell caches in the supplied workbook are empty.

Patient Monitor has the highest complaint total at 1,405. ECG Machine has the highest adverse-event total at 329, while Dialysis Machine has the highest adverse-event average at 24.83 per record. These comparisons identify concentrations in the dataset; they do not establish patient-level incidence or comparative device safety.

The dashboard contains five KPI cards, six charts and a device-type-by-status matrix. Its supporting sheet is named Pivot Table, but the summaries are implemented with formulas rather than native Excel PivotTable objects. The report explains the calculation logic, distinguishes the scope of different filters, and identifies changes needed for dependable refresh and interpretation.

Source: Tables!B2:B201, J2:K201; Medical Device Dashboard!B5:N7 and B50:G66; Pivot Table sheet and workbook object definitions.

1 Project objectives and dataset scope

The analytical purpose is to make a regulatory record set easier to review. A user can compare status distributions, locate groups with substantial recorded complaints or adverse events, examine recall flags, and inspect the relationship between device type and status. The workbook can also support a discussion of data completeness and the information needed for further investigation.

Questions addressed

The project asks how many records are Approved, which device types contribute the largest recorded event totals, how recall flags vary across groups, where records are geographically concentrated, and how the recorded approval-year profile differs from the subset marked Approved. It also makes UDI and postmarket-study labels available for supplemental analysis.

Dataset dimension	Observed coverage
Rows and fields	200 records and 15 fields
Device types	15
Class labels	4
Countries	10
Manufacturer labels	20
Framework and body labels	5 in each field
Recorded approval years	2019 to 2025
Last audit dates	24 January 2019 to 22 June 2024

Source: Tables!A1:O201. Coverage counts refer to distinct values in the supplied dataset.

Workbook organisation

Worksheet	Purpose
Medical Device Dashboard	Five KPIs, four dropdown controls, six charts and a status matrix
Tables	The source data range containing all 200 records
Pivot Table	Formula-based summaries, KPI calculations and a cross-tabulation

Each Device_ID is unique within the file. A record is the unit of analysis; the workbook does not contain units sold, installed population, patient exposure or sales value. Country counts therefore show the distribution of records. Complaint and adverse-event counts remain separate measures because the dataset does not define whether individual reports overlap.

The workbook does not identify a source-system publisher, reporting-period definition or status effective date. Regulatory labels are analysed as supplied. Their accuracy and applicability to a particular product and market require supporting documentation before operational or regulatory conclusions are drawn.

Source: Tables headers and values; workbook sheet, relationship and metadata inspection.

2 Data preparation and quality review

The source range Tables!A1:O201 already contains a complete set of 200 rows. The review checked identifiers, missing values, text consistency, numeric fields and audit-date parsing before calculating summaries. There is no separate raw-to-clean history, embedded Power Query transformation or record-rejection log in this workbook.

Quality check	Observed result
Device_ID uniqueness	200 unique identifiers; no duplicates
Duplicate full rows	None
Missing cells	None across the 15 populated fields
Leading or trailing spaces	None in the checked text values
Complaint counts	Integers from 0 to 150; no negative values
Adverse-event counts	Integers from 0 to 40; no negative values
Recorded approval years	Integers from 2019 to 2025
Audit dates	All 200 stored as text; all parse successfully
Audit date coverage	195 distinct dates from January 2019 to June 2024

Source: Independent checks of Tables!A2:O201. No source records were removed or changed for this report.

Date handling

Last_Audit_Date values use a consistent year-month-day string format. The analytical review parsed these strings as dates, while the original file retains text values. A future workbook update should store them as true Excel dates to support reliable date filtering, elapsed-time calculations and sorting.

For 110 records, the audit year precedes the value in Year_Approved. This does not automatically identify an error: the audit may relate to a manufacturer, an earlier assessment or a different approval event. The audit type, assessment scope and approval history should be recorded before using the sequence as a quality rule.

Definitions requiring clarification

Twenty-five records contain more adverse events than complaints. The file provides no rule stating that every adverse event must be included in the complaint count, so those values should be retained and investigated through field definitions. Similarly, every record has a Year_Approved value, including 144 records whose stored status is not Approved. These two fields need a clear explanation of their timing and meaning.

Completeness alone does not validate a classification or status. A useful next version would add the source reference, date of extraction, applicable market, status date and any rationale needed to interpret the recorded framework and class labels.

Source: Tables!I2:O201; comparison of audit dates, approval years, event fields and status labels.

3 KPI calculations and dashboard design

The five dashboard cards use formulas on the Pivot Table sheet or aggregate the source range directly. The following values were recalculated from Tables with all four filters set to All.

Dashboard label	Calculation	Full-data result
Total Devices	Count records in the four recognised status groups	200 records
Approved	Approved count ÷ total records	56 ÷ 200 = 28.0%
Recall Rate	Recall_Required Yes count ÷ total records	110 ÷ 200 = 55.0%
Avg Adverse Events	Total adverse events ÷ total records	3,898 ÷ 200 = 19.49
Total Complaints	Sum Complaints_Reported	15,488

Source: Medical Device Dashboard!B7, E7, H7, K7, N7; Pivot Table!V1:W7; Tables!J2:N201.

The Approved card is a percentage, although its label does not say so. The average adverse-event card is formatted to one decimal place and would display 19.5 after successful calculation; the underlying full-data mean is 19.49. Recall Rate is the share of records with a Yes flag, rather than a count of recall actions or units recalled.

Six dashboard views

View	Data summarised
Regulatory Status Distribution	Counts across four stored status labels
Complaints and Adverse Events by Device Type	Two event-count series across all 15 device types
Recall Required	Counts of Yes and No flags
Devices Sold by Country	Record counts and complaint totals across ten countries
Device Count by Class	Counts across four class labels
Devices Approved by Year	All records grouped by Year_Approved, regardless of stored status

Source: Dashboard chart definitions; Pivot Table!A1:T16. Chart labels are reproduced as used in the workbook.

Controls and summary logic

Country, Device Type, Device Class and Notified Body are selected in R5, R7, R9 and R11. All or a blank selection removes that condition. SUMPRODUCT performs conditional counts and totals, and IF handles selected calculation conditions. The device-type-by-status matrix is linked from Pivot Table! A20:F36 to the dashboard.

The views do not all apply the same filter scope. Type summaries retain all types when Device Type changes; country summaries retain all countries when Country changes; class summaries retain all classes when Device Class changes. Section 9 explains these differences and the calculation-cache limitations.

Source: Medical Device Dashboard!R5, R7, R9, R11; Pivot Table formulas and dashboard matrix references.

4 Regulatory status and recall flags

Rejected is the largest status group at 59 records, followed by Approved at 56, Pending Review at 50 and Under Investigation at 35. Approved records represent 28.00% of the dataset. Pending Review and Under Investigation together contain 85 records, or 42.50%, but describe different stored states.

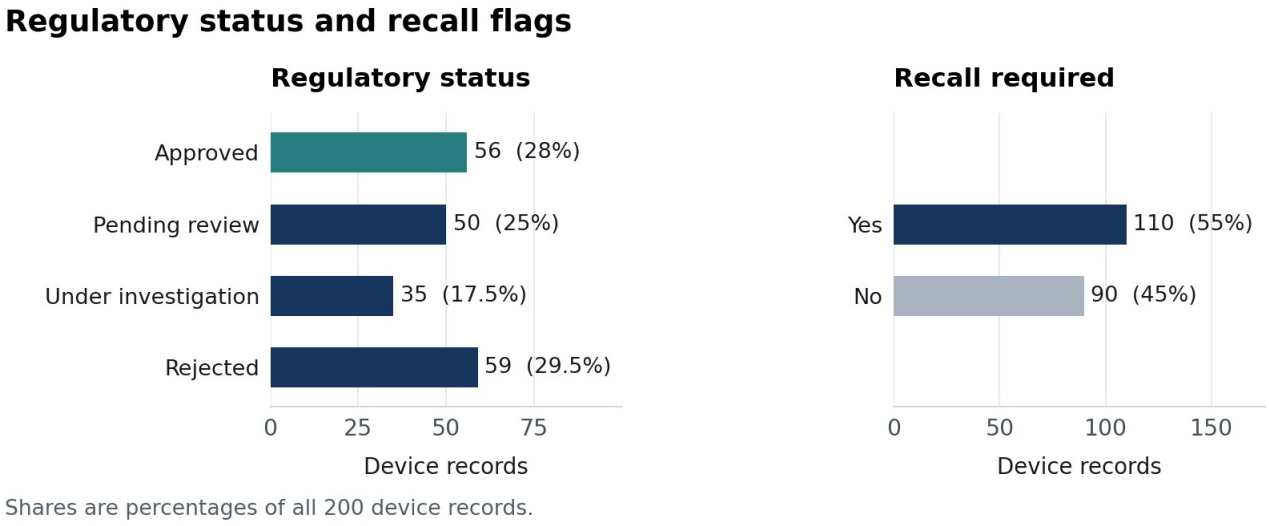


Figure 1 Stored regulatory statuses and recall flags for all 200 records

Source: Tables!L2:L201 and N2:N201; independently aggregated counts. Statuses and recall flags are separate attributes.

Recall flags within each status

Stored status	All records	Recall Yes	Within-group share
Approved	56	28	50.00%
Pending Review	50	28	56.00%
Rejected	59	32	54.24%
Under Investigation	35	22	62.86%

Source: Cross-tabulation of Tables!L2:L201 and N2:N201. All four groups together contain 110 Yes flags.

Twenty-eight Approved records also have Recall_Required set to Yes. Approval status alone therefore does not isolate the recall-review population. Likewise, 22 of the 35 Under Investigation records have a Yes flag. A review list should retain one row per Device_ID with both fields visible, so overlapping groups are not counted twice.

The 55.00% figure is a dataset flag proportion. It does not demonstrate that 55% of marketed units were recalled, that a recall has been completed, or that an equivalent proportion of patients experienced harm. Action dates, affected units, decision records and closure evidence would be needed to assess actual recall activity.

5 Complaints and adverse events by device type

The dataset contains 15,488 reported complaints and 3,898 reported adverse events. The averages are 77.44 complaints and 19.49 adverse events per record. The figures are sums of recorded counts; the workbook supplies no common event-reporting period or exposure denominator.

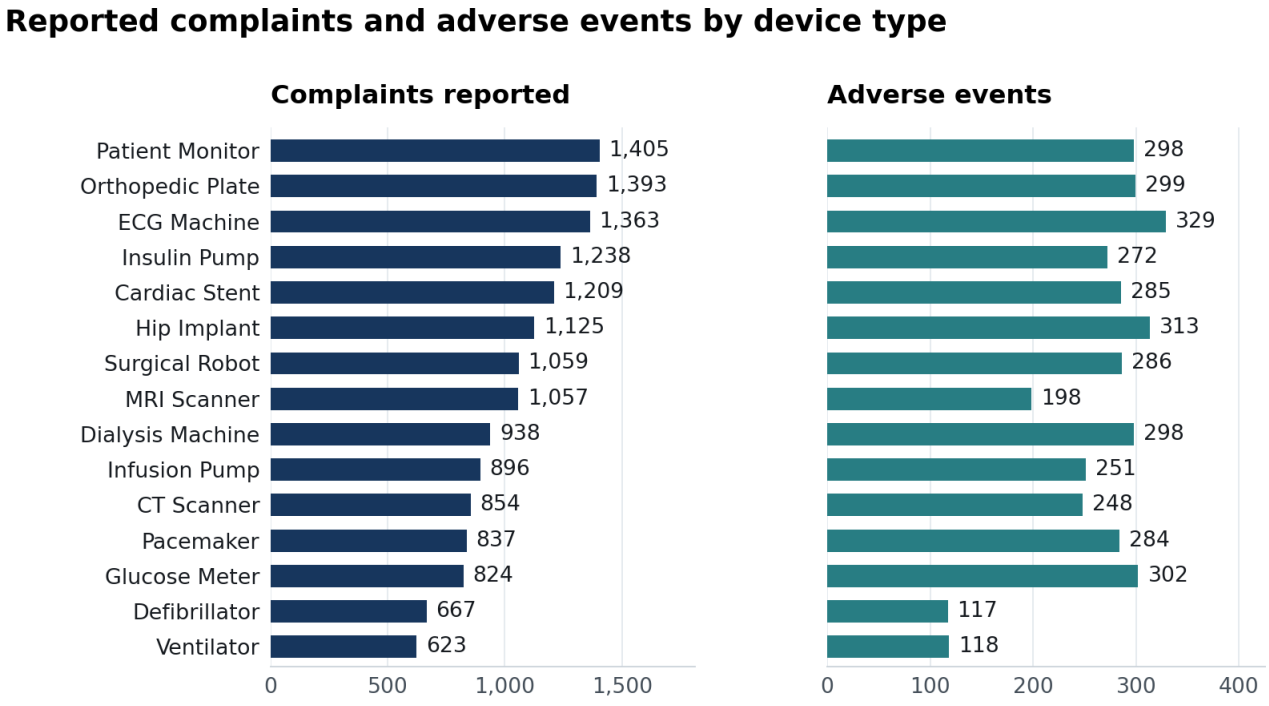


Figure 2 Complaint and adverse-event totals by device type

Source: Tables!B2:B201 and J2:K201. Device types ordered by complaint total; the two panels use different count scales.

Patient Monitor contributes the highest complaint total: 1,405 across 14 records. Orthopedic Plate follows with 1,393 across 16 records, then ECG Machine with 1,363 across 15. Together these three groups account for 4,161 complaints, or 26.87% of the total.

ECG Machine has the highest adverse-event total at 329. The largest average is in Dialysis Machine at 24.83 per record, compared with the overall 19.49. Patient Monitor also has the highest complaints per record at 100.36. These comparisons show why totals should be considered together with the number of records in each group.

Recall flags have a different distribution: Orthopedic Plate has the highest flagged count at 12 of 16 records, while Hip Implant has the highest within-type share at 11 of 14, or 78.57%. The groups are small, and differences in event counts or flags do not establish differences in product safety without exposure, severity, case validation and reporting-period information.

Source: Independent grouping of Tables!B2:B201 and J2:L201. Counts and per-record means are descriptive comparisons.

6 Geographic coverage and device classes

Canada and the USA each account for 25 records, or 12.50% of the dataset. Germany and the UK follow with 22 records each. These are counts of records labelled with each country; no physical sales quantity is supplied.

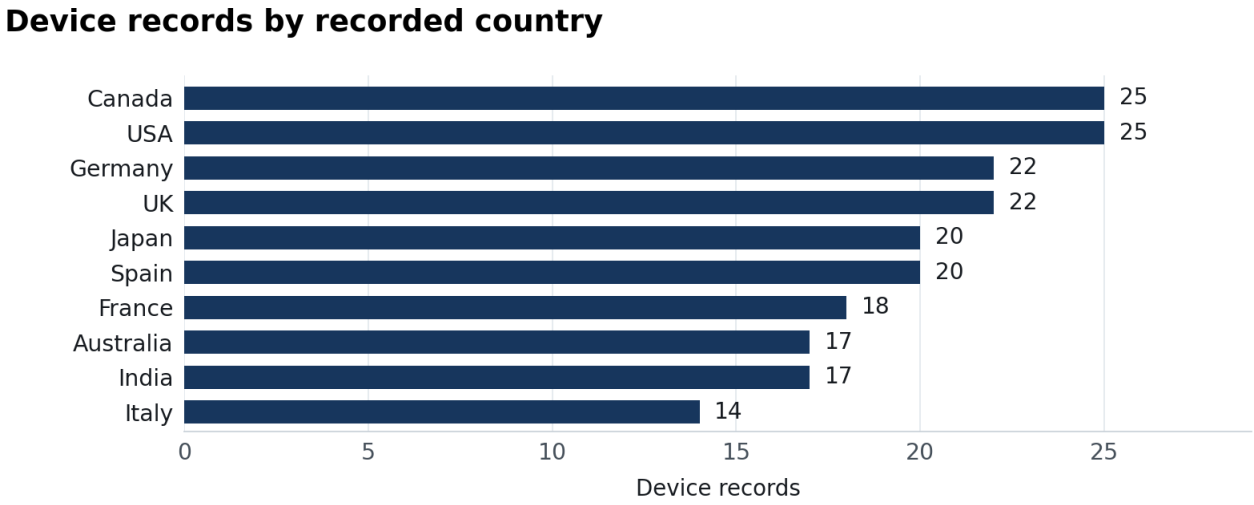


Figure 3 Distribution of device records across the ten country labels

Source: Tables!D2:D201; intended country summary at Pivot Table!L1:N11.

Canada has the largest country complaint total at 2,034 and adverse-event total at 539. Canada, the UK and the USA together contribute 5,702 complaints, or 36.82% of the total. Germany has the highest adverse-event average at 23.64 per record across 22 records. Country comparisons are affected by the mix of device records and their reporting histories.

Stored class profile

Class label	Records	Complaints	Adverse events	Mean events
Class I	57	5,282	1,079	18.93
Class IIa	44	3,145	761	17.30
Class IIb	49	3,416	1,022	20.86
Class III	50	3,645	1,036	20.72

Source: Tables!C2:C201 and J2:K201; independently calculated totals and means.

Class I has the most records and the highest complaint total. Class IIb has the highest mean adverse-event count at 20.86, slightly above Class III at 20.72. These results describe the supplied class labels. The workbook spans multiple framework labels and does not document classification rules, intended use or the market-specific rationale needed to validate each assignment.

7 Recorded approval years and audit dates

Year_Approved contains a year for every record, including records marked Pending Review, Rejected and Under Investigation. The workbook chart counts all records with each year value. The report separates that overall profile from the subset with Regulatory_Status equal to Approved.

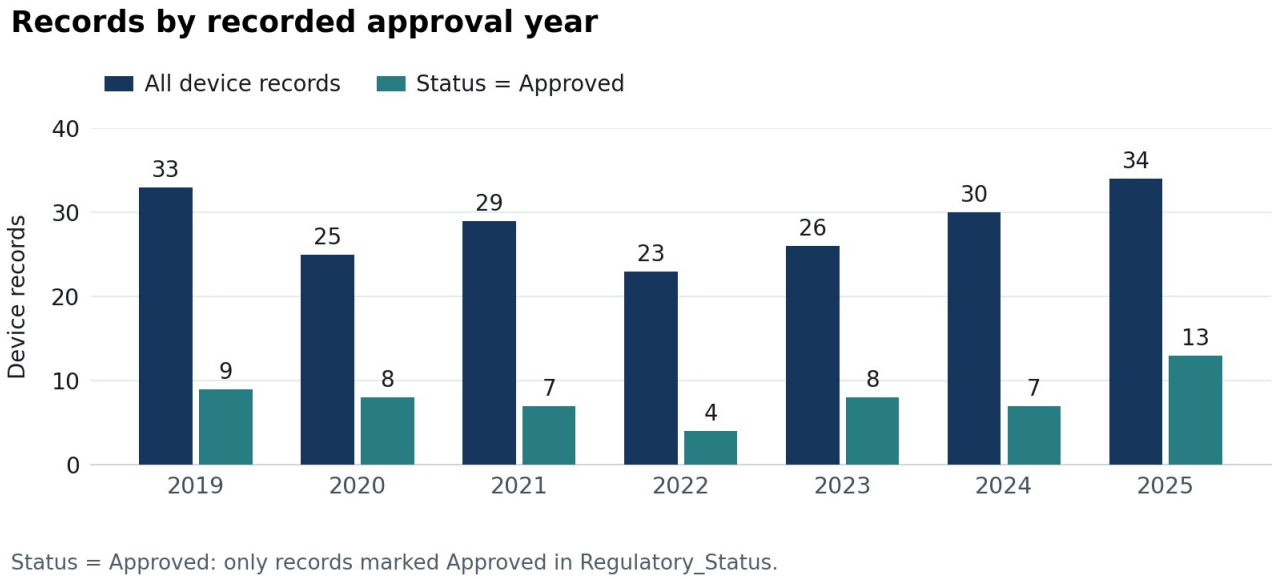


Figure 4 All records and the Approved subset by recorded approval year

Source: Tables!I2:I201 and N2:N201. Approved subset is recalculated for this report; the workbook year chart uses all statuses.

Recorded year	All records	Marked Approved	Other statuses
2019	33	9	24
2020	25	8	17
2021	29	7	22
2022	23	4	19
2023	26	8	18
2024	30	7	23
2025	34	13	21

The largest recorded-year group is 2025 with 34 records, of which 13 are marked Approved. The smallest is 2022 with 23 records, of which four are marked Approved. These are distributions of stored year and status values; they do not establish the number of regulatory decisions issued in each calendar year.

Audit information

The latest recorded audit date is 22 June 2024 and the earliest is 24 January 2019. No record has an audit date in 2025 or 2026. The workbook does not state the date of extraction, required audit frequency, audit type or next due date. Consequently, this report does not classify records as overdue solely from the age of their audit dates.

Source: Tables!I2:I201, N2:O201; Pivot Table!S1:T8. Audit dates parsed from the source text strings.

8 UDI postmarket and framework information

UDI_Assigned is Yes for 101 records and No for 99. PostMarket_Study is Required for 106 records and Not Required for 94. These fields describe recorded flags; the workbook does not contain the UDI itself, applicability rationale, study dates or study completion status.

Review population	Records	Reason to retain the distinction
Approved and UDI No	21	Approval status and identification flag are separate
Recall Yes and UDI No	53	Potential traceability review population
Recall Yes and study Required	56	Two flags needing separate supporting records
UDI No and study Required	49	Overlaps other review groups

Source: Cross-tabulations of Tables!H2:H201 and L2:N201. Groups overlap and are not additive.

Recorded framework labels

Framework label as supplied	Records	Share	Marked Approved
EU MDR 2017/745	46	23.00%	13
FDA 21 CFR 820	46	23.00%	14
TGA Australia	39	19.50%	10
UK MDR 2002	38	19.00%	9
Health Canada CMDR	31	15.50%	10

Source: Tables!F2:F201 and N2:N201. Labels retained verbatim from the dataset.

EU MDR 2017/745 and FDA 21 CFR 820 each occur in 46 records. Country_Sold and Regulation_Framework are independent fields in this file. Their relationship should be verified against the product and market documentation before using either field to determine a regulatory obligation or an approval route.

Named body profile

Body label	Records	Marked Approved	Recall Yes
DEKRA	44	13	21
Intertek	43	11	27
SGS	39	10	20
BSI	37	11	22
TUV SUD	37	11	20

Source: Tables!G2:G201 and L2:N201. Counts refer to labels stored in the project file.

DEKRA has the most body-labelled records at 44. These counts can help route a document review, but they do not measure a named organisation’s performance or validate its role for any particular market. No certificate identifiers, scope statements or assessment dates are supplied.

9 Formula verification and implementation limits

The workbook has 261 formula cells and six charts. No native Excel PivotTable definitions or structured Excel table objects are present. The source occupies a worksheet named Tables, and the summary worksheet is named Pivot Table. The report uses precise descriptions of those implemented structures.

Saved calculation results

All 261 formula cells lack saved result values. The six charts retain cached series, and all 67 stored numeric points match independently calculated All-selection summaries. Chart caches are separate from formula-cell results. Report totals were independently calculated from the 200 source records; the saved KPI cells provide no calculated values.

Different filter scopes

View or calculation	Filter condition omitted
Device-type totals	Device Type selection in R7
Device-type-by-status matrix	Device Type selection in R7
Country totals	Country selection in R5
Class totals and class averages	Device Class selection in R9
Global KPIs, status and recall summaries	Designed to apply all four selections

Source: Pivot Table!D2:G16, L2:R11 and A21:F36; Medical Device Dashboard!R5, R7, R9, R11; KPI formulas.

This design retains comparison categories when the user selects that same dimension. For example, selecting Canada changes the global KPIs, while the country panel retains the other country categories. The dashboard subtitle says the dropdowns filter all charts, which is broader than the implemented scope. The behaviour should either be made consistent or explained beside the affected charts.

Labels and future refresh

The Approved card should be labelled Approved Percentage, and Total Devices should make the record unit explicit. The year chart should identify that it groups all records by Year_Approved unless an Approved-status condition is added. The country view includes record counts and complaint totals, so both series and their different meanings need clear labels.

Source formulas stop at row 201, and dropdown choices and chart categories are fixed lists. New rows and category values require range maintenance. The current total-record KPI is the sum of four recognised status groups, so an unexpected future status could be excluded. A direct validated Device_ID count would provide an independent total.

Source: Medical Device Dashboard!B5:N7, B30, K30 and filter validations; Pivot Table!B2:B6 and S2:T8; source ranges.

Calculation status

Formula review shows that a selection with no matching records returns zero for percentages and means, although these measures are undefined without a denominator. No interactive spreadsheet calculation test was performed. Recalculate and save the workbook in a compatible application, then verify filtered results against the source before recurring use.

10 Recommendations and project outcome

Make the dashboard unambiguous

Rename KPI labels to state whether they represent a count, percentage or mean. Describe Recall Rate as the share of records flagged Yes. Clarify the approval-year chart's status scope and the country panel's record-count basis. Show the active filters and record denominator clearly, and explain any panel that deliberately keeps comparison categories visible.

Strengthen calculation and refresh controls

Recalculate the workbook in a compatible spreadsheet application and save the resulting values. Reconcile the five KPI cards with the source totals and test a country, a device type, a combined selection and a selection with no matching records. Replace fixed ranges with expanding sources and maintain the dropdown lists when new categories arrive.

Use a direct Device_ID count as a control, with a check that the status summaries add to the same total. Represent an undefined percentage or average as No matching records rather than a measured zero. Retain a dated refresh log that records input rows, excluded rows, calculation checks and the reporting cutoff.

Create traceable review lists

Build a list of the 110 Recall Required records containing Device_ID, stored status, country, manufacturer, UDI flag and the supporting action reference. Keep the 53 records also marked UDI No visible within that list. Use the 50 Pending Review and 35 Under Investigation records as distinct status groups, rather than combining them into an undifferentiated compliance-failure count.

Review Patient Monitor, Orthopedic Plate and ECG Machine complaint records because they contribute 26.87% of complaints. Compare records by exposure and reporting period before setting a risk priority. The 106 PostMarket Study Required records need the study owner, due date and progress information before a completion or overdue measure can be calculated.

Add the evidence needed for stronger conclusions

Useful additions include source-system provenance, dates for each status and event, product-model identifiers, affected units or exposure, event severity, investigation outcome, recall action dates, certificate references, and audit type and next due date. Definitions should explain whether complaint and adverse-event counts overlap and what Year_Approved represents for each status.

Project outcome

The project demonstrates source-data organisation, conditional aggregation, percentage and mean calculations, dropdown controls, cross-tabulation and visual comparison in Excel. It gives the user a compact view of a 200-record regulatory dataset and supports a focused discussion of review populations. The analysis also identifies specific calculation, naming, filtering and provenance issues to address before relying on the dashboard as a recurring operational report.

Source: Recommendations follow the findings in Sections 2-9. Source totals and reference ranges are indexed in Section 12.

11 Data dictionary

The source range contains the following 15 fields. Definitions describe their use in the workbook and the interpretation applied in this report.

Field	Col	Meaning
Device_ID	A	Unique identifier for a device record
Device_Type	B	Equipment-type label
Device_Class	C	Stored class label; classification basis not supplied
Country_Sold	D	Country label associated with the record
Manufacturer	E	Stored manufacturer name
Regulation_Framework	F	Framework label assigned in the dataset
Notified_Body	G	Named body label recorded for the device
UDI_Assigned	H	Yes or No identification-assignment flag
Year_Approved	I	Recorded year; populated for every status
Complaints_Reported	J	Integer complaint count
Adverse_Events	K	Integer adverse-event count
Recall_Required	L	Yes or No recall-requirement flag
PostMarket_Study	M	Required or Not Required study flag
Regulatory_Status	N	Approved, Pending Review, Rejected or Under Investigation
Last_Audit_Date	O	Text date of the recorded last audit

Source: Tables!A1:O201. The workbook contains no separate data-dictionary worksheet.

Units and denominator conventions

Record percentages use 200 as the denominator for the full dataset, or the stated group size for a within-group comparison. A mean event count is the sum of recorded events divided by the relevant number of records. It is not a patient rate or a rate per unit sold. There is no financial measure or device-quantity measure in this source file.

A Yes flag records the condition named by its field. Recall_Required does not establish a completed recall, UDI_Assigned does not verify the actual identifier, and PostMarket_Study does not record whether a study has started or finished. The report retains those differences when describing review groups.

Date and status conventions

The recorded approval-year range is 2019-2025. Last audit dates span 2019-2024 and were parsed for analysis from year-month-day text. The report date identifies when this report was prepared; it is not a verified status date or a date through which complaints and adverse events have been collected.

12 Status matrix and evidence references

Device type	Approved	Pending	Rejected	Investigating	Total
CT Scanner	3	2	5	2	12
Cardiac Stent	5	4	4	2	15
Defibrillator	3	0	4	0	7
Dialysis Machine	1	4	2	5	12
ECG Machine	2	5	2	6	15
Glucose Meter	5	1	3	4	13
Hip Implant	3	4	4	3	14
Infusion Pump	4	2	7	1	14
Insulin Pump	2	7	6	3	18
MRI Scanner	4	2	6	0	12
Orthopedic Plate	4	4	5	3	16
Pacemaker	4	4	2	3	13
Patient Monitor	7	3	2	2	14
Surgical Robot	6	4	5	0	15
Ventilator	3	4	2	1	10
Total	56	50	59	35	200

Source: Recomputed from Tables!B2:B201 and N2:N201. Column order follows the dashboard matrix: Approved, Pending Review, Rejected, Under Investigation.

Workbook reference index

Evidence	Worksheet location
Source dataset	Tables!A1:O201
KPI cards and controls	Medical Device Dashboard!B5:N7; R5, R7, R9, R11
Status and recall summaries	Pivot Table!A1:B6 and I1:J3
Device-type event summaries	Pivot Table!D1:G16
Country and class summaries	Pivot Table!L1:N11 and P1:R5
Year profile and KPI helpers	Pivot Table!S1:T8 and V1:W7
Device-type-by-status matrix	Pivot Table!A20:F36; Medical Device Dashboard!B50:G66

Analytical method

The review independently aggregated the source records, checked data completeness and date parsing, inspected workbook objects, and traced KPI and chart references. Report charts were drawn from the recomputed source totals. Formula inspection identifies intended dependencies and scope differences; it does not substitute for a native Excel calculation test. The source workbook was analysed without modification.

Primary source: Shraddha_Dashboard_FINAL2.xlsx. Project attribution is recorded in Medical Device Dashboard! B3. All numerical findings in this report are derived from that file.